

PCOS Care AI

AI-Based PCOS Risk Assessment & Personalized Care System

A plain-English guide to what this project is, how it is built, and the tools behind it
— written for someone with no technical background.

Important: PCOS Care AI is an awareness and decision-support tool. It gives a preliminary risk estimate only and is **not** a medical diagnosis. Always consult a qualified doctor.

1. What is this project?

PCOS (Polycystic Ovary Syndrome) is a common hormonal condition in women. Many people do not realise they may have it. **PCOS Care AI** is a website where a person answers a few simple questions about their health and lifestyle. The system then uses **Artificial Intelligence (AI)** to estimate how likely it is that they show PCOS-related patterns, and gives friendly lifestyle suggestions.

Think of it like this

Imagine a very well-read health assistant. You tell it about yourself, it compares your answers to patterns it has learned from thousands of examples, and it says "your answers look low / medium / high risk — here are some healthy habits, and please see a doctor if needed." It never replaces the doctor; it just helps you decide to go.

2. What can it do? (Features)

The system has three types of users, each with their own screen:

Patient

- Create an account and log in.
- Fill in a short health & lifestyle questionnaire.
- Instantly see a risk result (Low / Moderate / High) with a percentage.
- Get personalized, non-medical lifestyle recommendations.
- Download the result as a PDF report (including all their answers).
- See their past assessments and any prescriptions their doctor uploaded.

Doctor

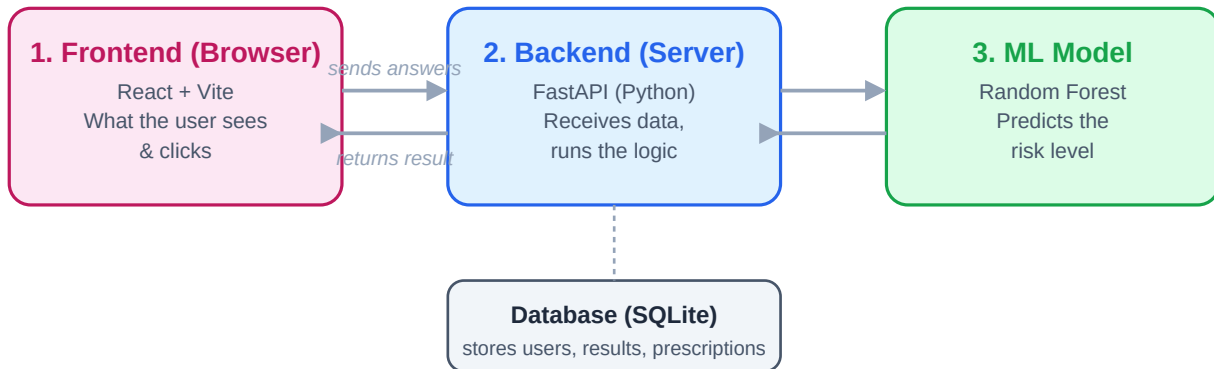
- Log in to a review dashboard.
- See every patient's assessment, filter by risk level.
- Download a patient's PDF report.
- Upload a prescription file (PDF/image) for a patient.

Admin

- Log in to an analytics dashboard.
- See how accurate the AI model is (accuracy, precision, recall, etc.).
- See charts: which factors matter most, and usage statistics.
- Re-train the model with one click.

3. How is it built? (Architecture)

The app has three main parts that talk to each other. When a patient clicks "Get my result", their answers travel from the browser to the server, the AI model makes a prediction, and the result travels back — all in about a second.



The three parts explained simply

- **Frontend** — the part you see in the web browser (pages, buttons, forms). Built with React. It is like the dashboard and steering wheel of a car.
- **Backend** — the hidden "engine" running on a server. Built with FastAPI (Python). It receives your answers, checks your login, talks to the AI model, and saves data.
- **Machine Learning model** — the "brain" that has learned from data and makes the actual risk prediction (a Random Forest).
- **Database** — the "filing cabinet" that remembers users, results and uploaded prescriptions (SQLite).

4. How does the AI make a decision?

The model is a **Random Forest**. Picture a room full of decision-makers, each a "decision tree" that asks yes/no questions like "Is the cycle longer than 35 days?" or "Is the BMI above 25?". Each tree votes on the outcome, and the forest combines all the votes into one confident answer. Using many trees together is more reliable than trusting just one.

How it learned

- It was **trained** on a dataset of examples (health details + whether PCOS patterns were present).
- 80% of the data was used to teach it; 20% was kept hidden to test it fairly afterwards.
- Its report card (accuracy, precision, recall, F1, ROC-AUC) is shown on the Admin dashboard.

Note on the data: the project ships with a realistic **synthetic** (computer-generated) dataset so it runs out of the box. For real-world use it should be re-trained on a validated clinical dataset.

5. Technology stack (the tools used)

Each tool below was chosen for a specific job. You do not need to memorise them — this is a reference.

Layer	Tool	What it does (in plain words)
Frontend	React + Vite	Builds the interactive web pages quickly.
Styling	Tailwind CSS	Makes the pages look clean and modern.
Charts	Recharts	Draws the dashboard graphs and charts.
Backend	FastAPI (Python)	The server engine that runs the logic.
Language	Python	The programming language of the backend + AI.
Machine Learning	scikit-learn	The AI toolkit that builds the model.
Algorithm	Random Forest	The specific AI method used to predict.
Data handling	Pandas + NumPy	Organise and crunch the numbers.
Database	SQLite	Stores users, results and files.
PDF reports	Browser print	Turns a result into a downloadable PDF.
Passwords	PBKDF2 hashing	Stores passwords safely (scrambled).

Supporting tools

Tool	Purpose
Node.js + npm	Runs and installs the frontend.
Uvicorn	Runs the FastAPI backend server.
joblib	Saves and loads the trained AI model.
Docker	Packages the whole app to run anywhere.
Render (free)	Puts the app online (hosting / deployment).
Git + GitHub	Stores and shares the source code.

6. What the results mean

Result	Meaning	Suggested action
Low	Few risk factors present.	Keep up healthy habits.
Moderate	Some risk factors present.	Consider a check-up.
High	Several risk factors present.	See a doctor soon.

The colours (green, amber, red) and a percentage make the result easy to understand at a glance.

7. How to run it (very simple)

You only need two free programs installed:

- **Python** (version 3.10 or newer) — runs the backend and AI.
- **Node.js** (version 18 or newer) — runs the website.

Then start two servers (each in its own window):

- **Backend:** run `run_backend.ps1` → starts the engine at `localhost:8000`.
- **Frontend:** run `run_frontend.ps1` → starts the website at `localhost:5173`.
- Open **`http://localhost:5173`** in a browser — done!

Demo logins

Role	Email	Password
Doctor	doctor@pcos.ai	doctor123
Admin	admin@pcos.ai	admin123
Patient	register your own	-

8. Mini dictionary (jargon buster)

Word	What it really means
AI / Machine Learning	Software that learns patterns from examples instead of being told every rule.
Model	The trained 'brain' that makes predictions.
Frontend	The part of the app you see and click in the browser.
Backend	The behind-the-scenes server that does the work.
API	The messenger that lets the frontend and backend talk.
Database	Where information is stored and remembered.
BMI	Body Mass Index — a weight-vs-height health number.
Training	Teaching the model using past examples.
Accuracy	How often the model gets its prediction right.
Deployment	Putting the finished app online for others to use.
Random Forest	An AI method that combines many decision trees.

PCOS Care AI — combining machine learning, web technology and data analytics into an accessible health-awareness tool. Not a medical device.